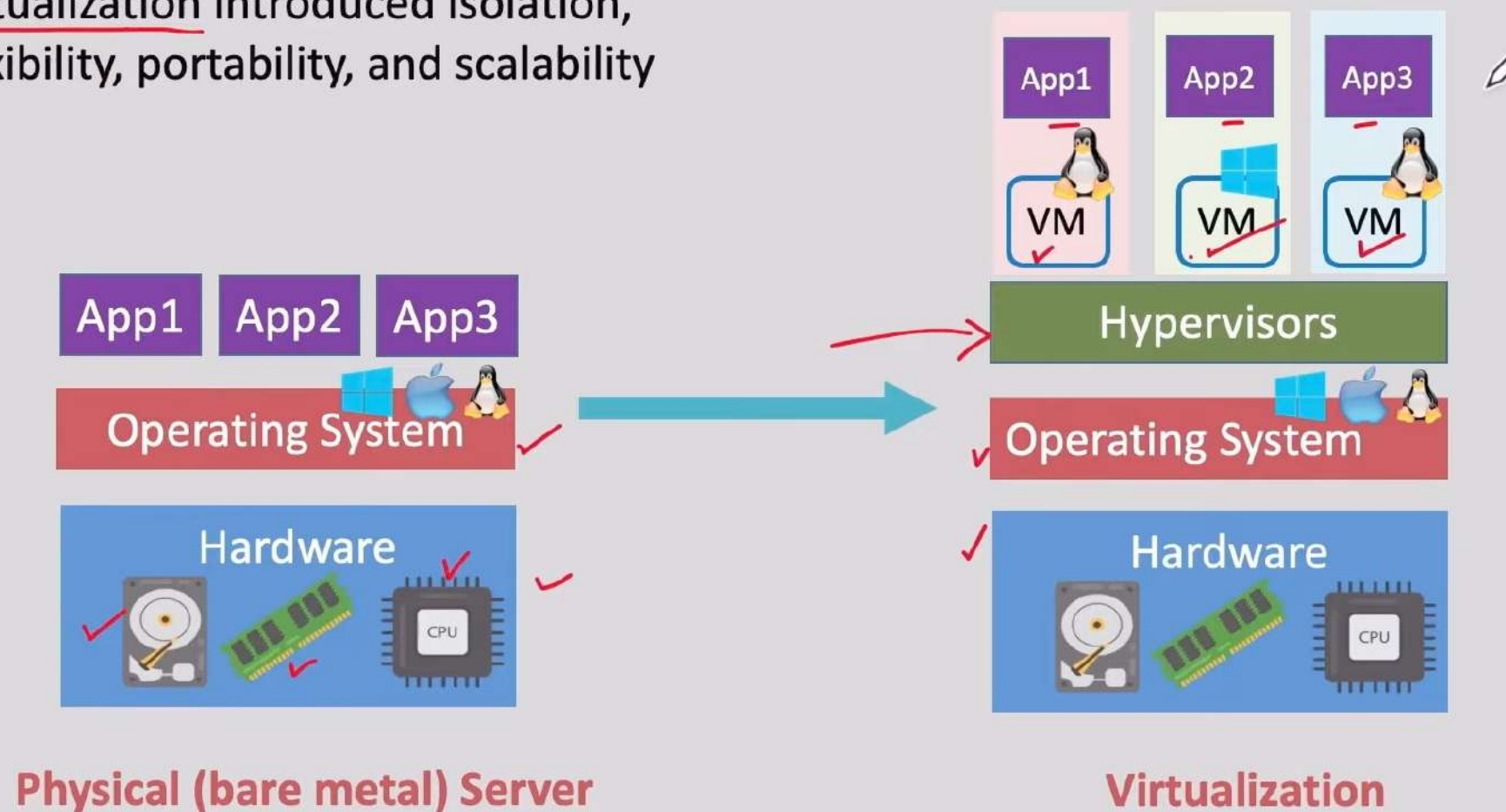


Introduction to Virtualization

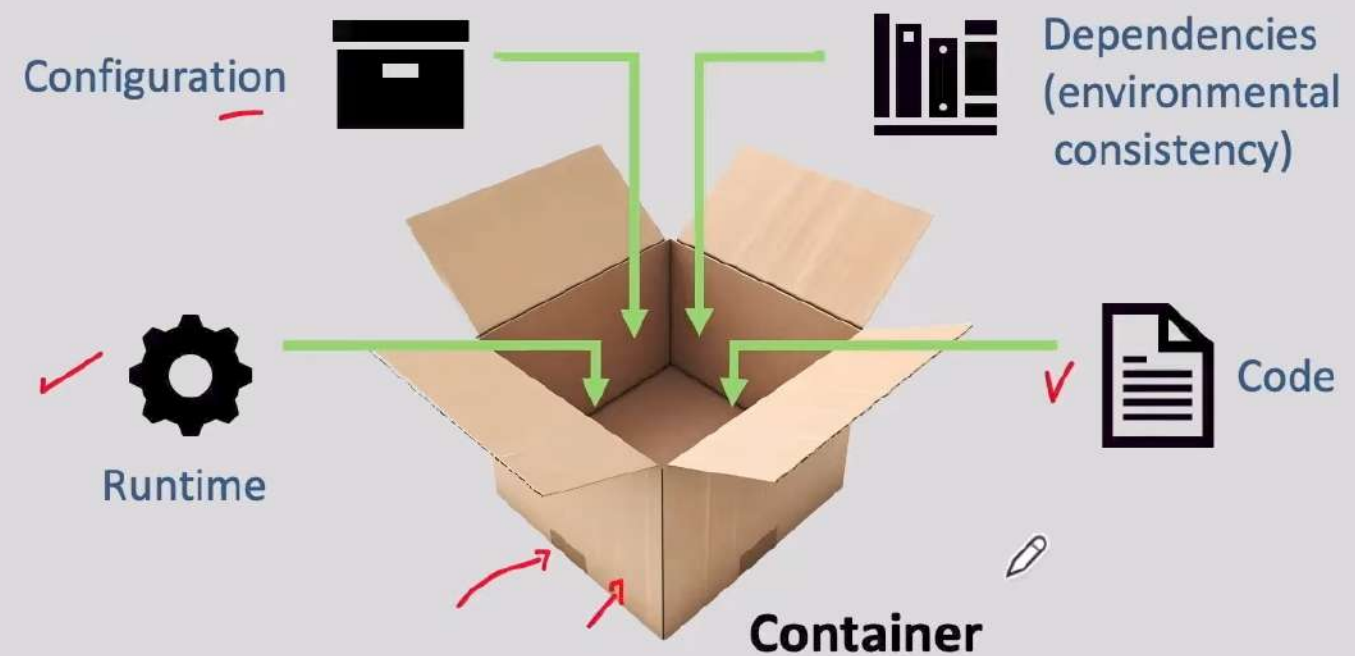
Virtualization introduced isolation, flexibility, portability, and scalability



Containers Overview

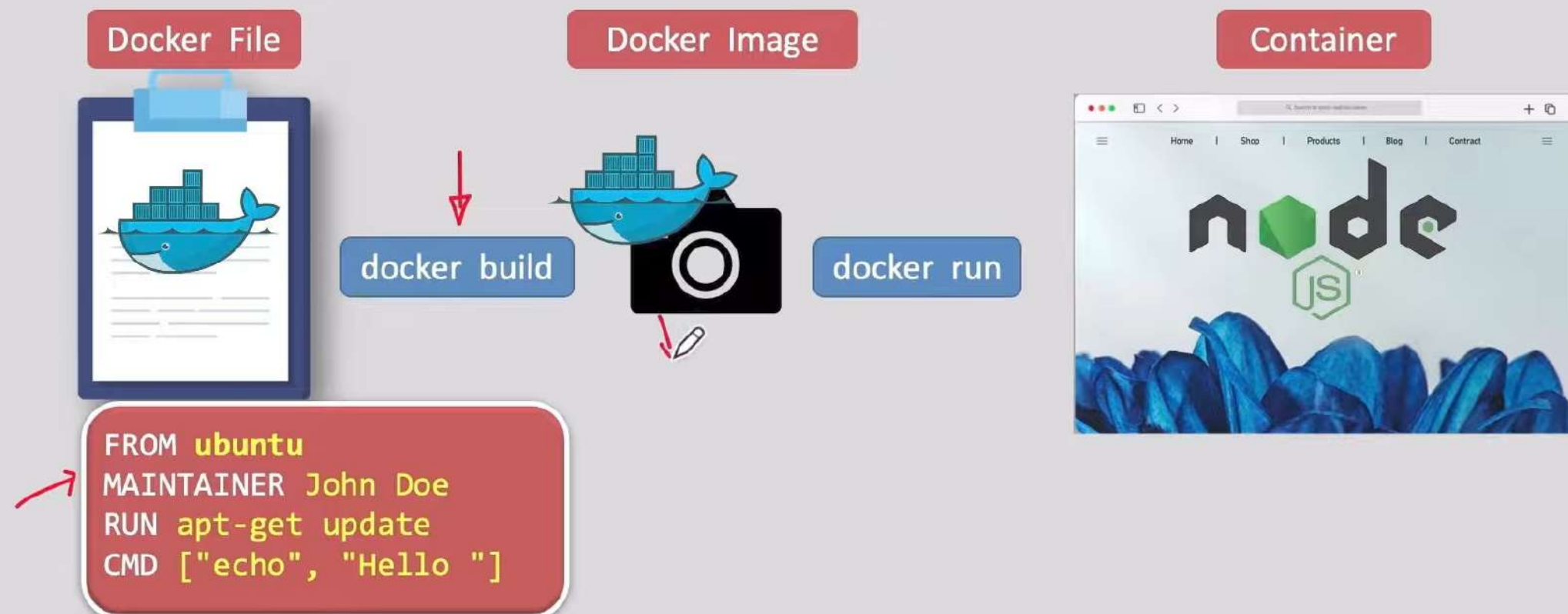


A **container** is a lightweight, isolated process running on a shared operating system kernel, packaged with everything it needs to run, including code, libraries, environment variables, runtime, and configuration files.

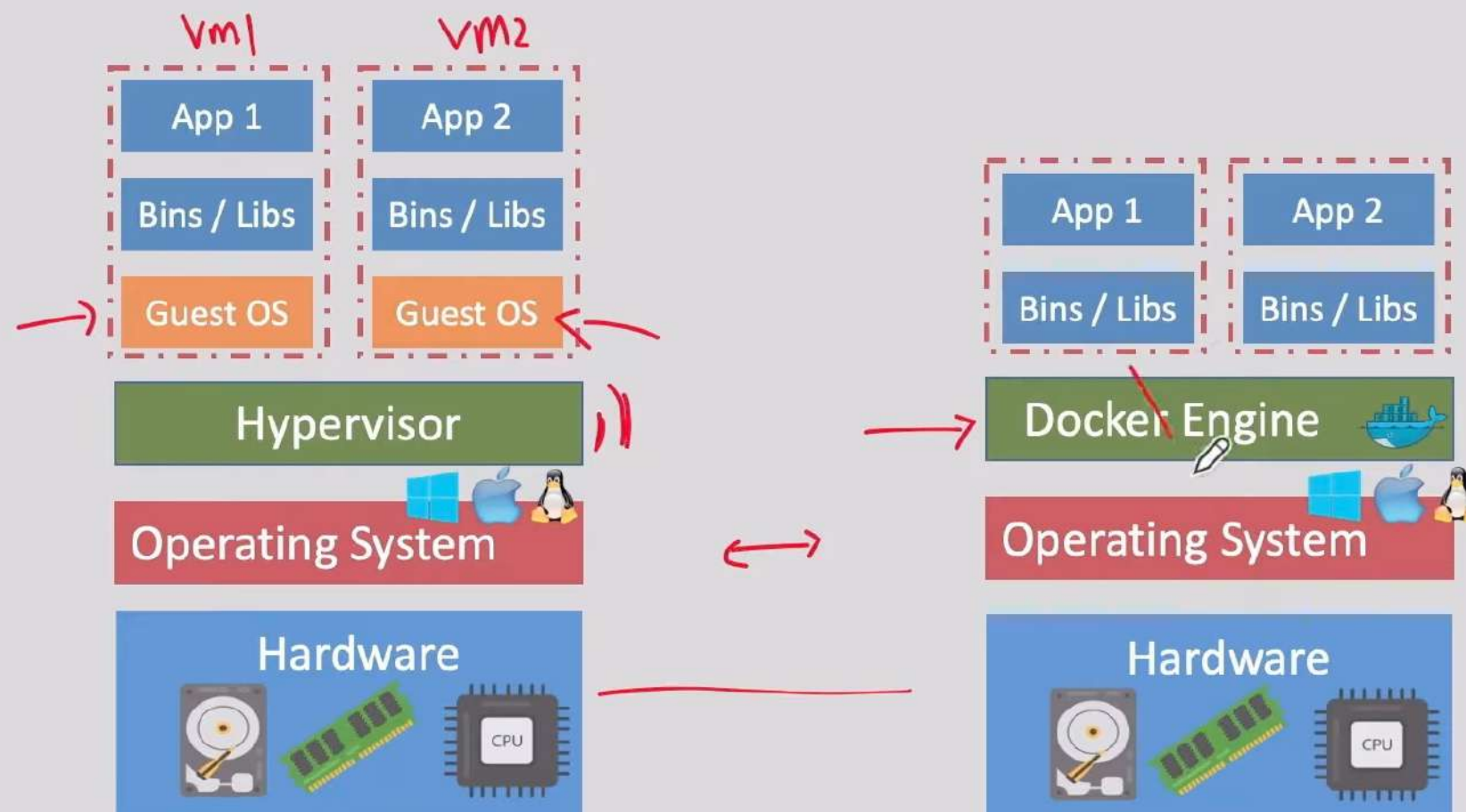


Docker Containers – From a Dockerfile to an Application

Starting with a Dockerfile, we can build a container image that can be used to consistently run an application across different environments.

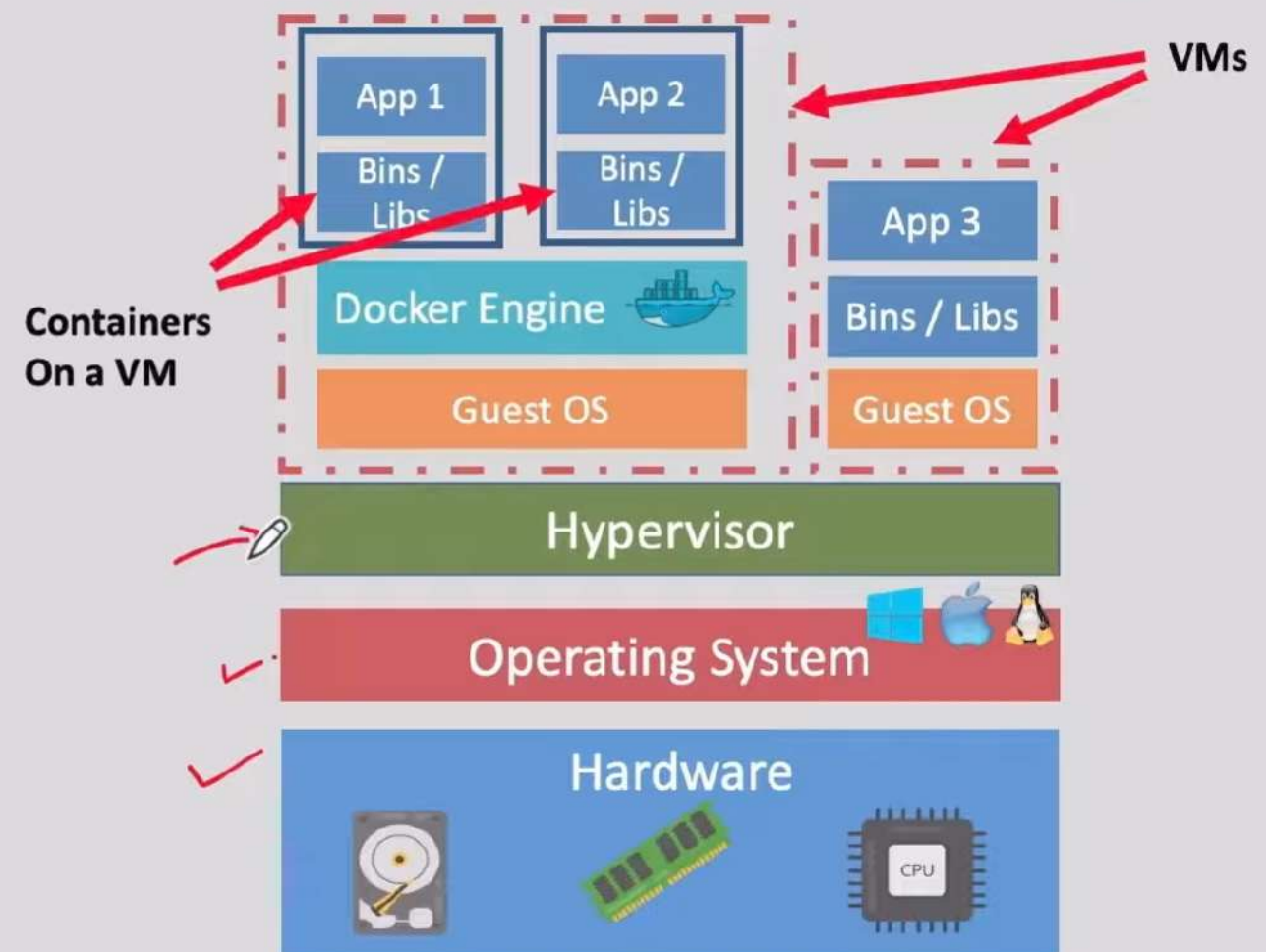


Virtual Machines (VMs) vs. Containers



Virtual Machines (VMs) vs. Containers

We can run containers on Virtual Machines

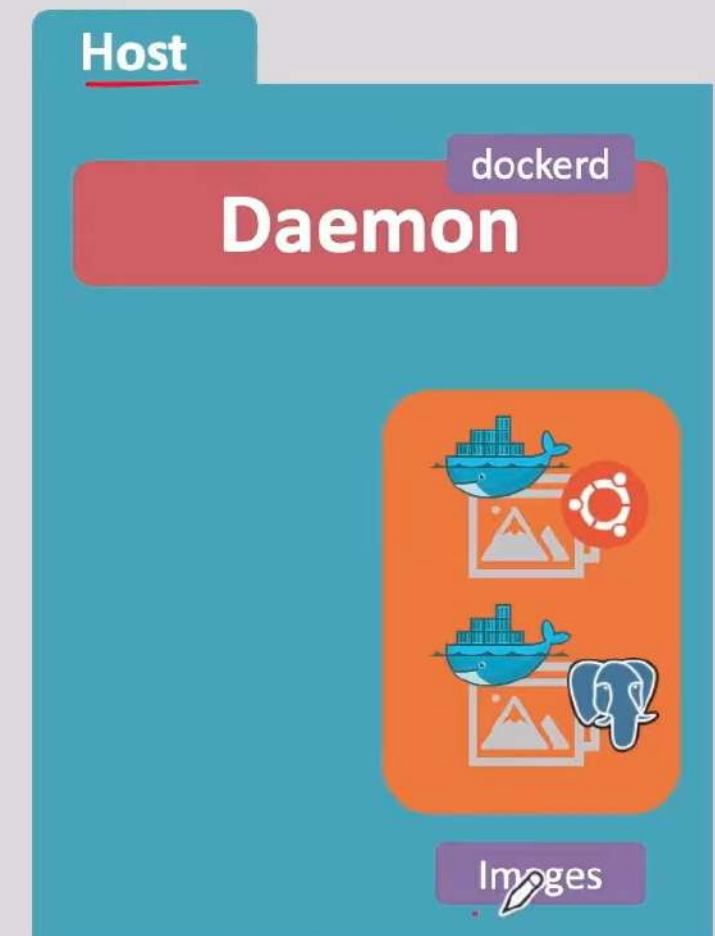


Docker Architecture – Docker Images



Images are read-only binary templates used to build containers

- They contain the application code, libraries, and dependencies required to run an application



Docker Architecture – Docker Containers



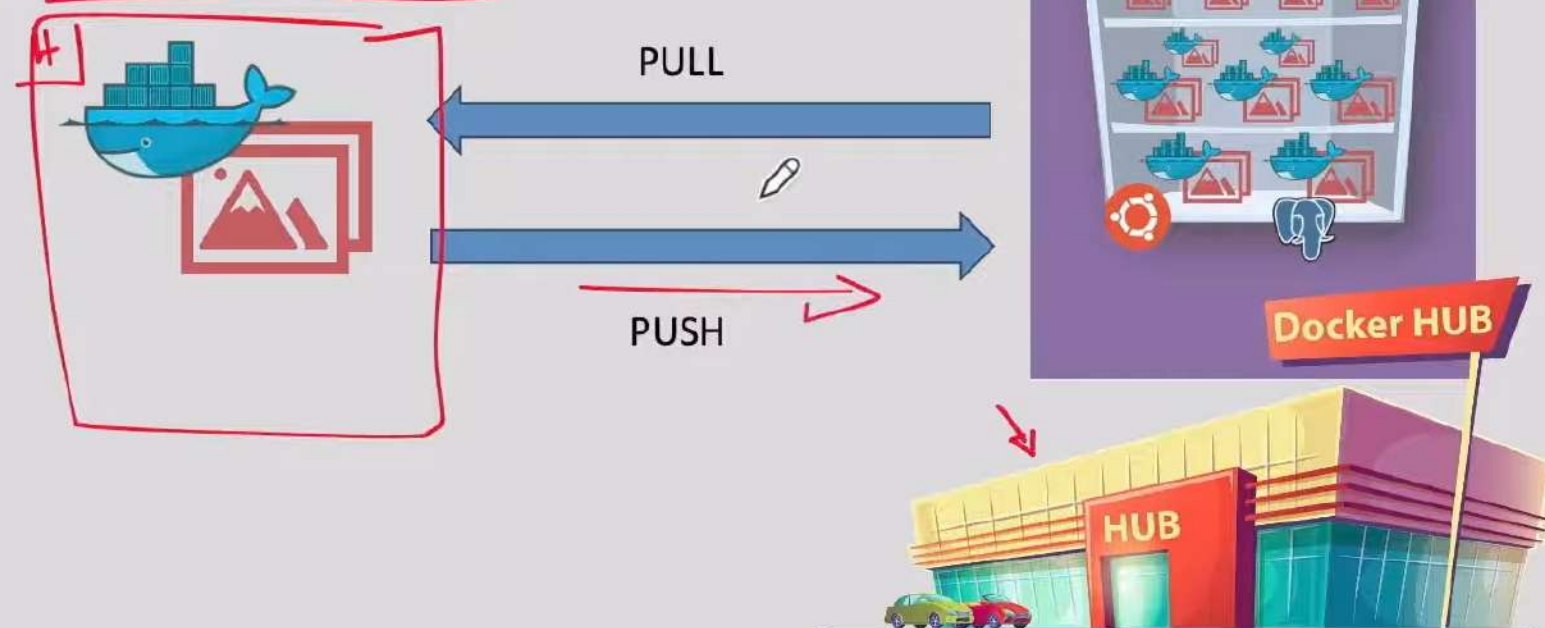
Containers are encapsulated environments in which you run applications



Docker Architecture – Docker Registries



- A Docker registry stores Docker images
- Docker Hub is the public registry that anyone can use
- When needed, the required images can be pulled from the configured registry
- Users can push their created images to the configured registry



Docker Architecture – Connecting the Dots

